

Affective Valence and Temporal Framing: A Unified Generative Model of the Temporal Structure of Emotion

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Abstract

Emotions are temporally structured: regret concerns what has happened, anxiety what will happen, joy what is happening now. Three strands of the active-inference literature each formalise one temporal face of affect: backward valence as the negative rate of change of variational free energy (Joffily and Coricelli, 2013), present valence as reward prediction error (Pattisapu et al., 2025), and forward valence as the affective charge of policy revision (Hesp et al., 2021). Yet each alone is *insufficient*: none spans the full temporal range, unifies these quantities into a taxonomy of emotion, or models how an agent actively selects the temporal horizon that dominates its own affect. We develop a single factored generative architecture in which a hidden *temporal-frame* state (past/present/future) sets the precision of the three channels and of the horizon terms of expected free energy, so that the inferred frame changes both what the agent feels and how it scores its next action. The unification is structural: one architecture with shared channel definitions, instantiated with a task-specific generative model and parameters fit per dataset, not a single jointly fitted model. We motivate the architecture from the psychology of mental time travel, the graded, cost-bounded regulation of how deeply memories and plans are unrolled, and show that a single *depth regulator* with two temporal faces reproduces the classical split between rumination (past-directed) and worry (future-directed). Affect is a *readout layer* over a task-specific generative model, and the three channels are general operators on its inference dynamics. Empirically, driven through real experience-sampling sequences from two independent samples, the full model predicts next-moment affect out of sample better than linear autoregressive baselines, roughly doubling one-step skill where affect carries volatile dynamics (held-out $R^2 = 0.19$ vs 0.09, participant-bootstrap CI on the difference [0.09, 0.12]); on a second, independent sample the lead does not reverse (ahead at two and three steps, parity at one). Ablations locate the gain in the model’s valence transition dynamics: it survives removing the persistence prior, and it is unchanged when the temporal frame is made inert or clamped, so the frame’s causal role is established in the simulated phenotypes (recall collapse, future fixation under stress, a diathesis-stress mood attractor, all robust across a sweep of the hand-set constants) rather than in affect prediction. On a static gamble the readout recovers the standard happiness equation ($n = 14,803$ held out) where each predecessor channel alone falls short, reaching that level without injecting expected value as a regressor (the forward channel supplies it by construction, a +30% gain over a reward-prediction-error channel). A counterfactual regret term reproduces part of the human tendency to switch after a loss when the foregone option was better ($t = 10.4$, replicated on a second dataset): a factual learner generates less than half of that contrast, counterfactual mechanisms most of it, at no gain in held-out choice likelihood. A mood layer’s diathesis-stress prediction holds in independent data, where a baseline vulnerability trait both lowers mood and amplifies affective stress-reactivity (interaction $t = 4.2$), the same conjunction that in simulation collapses into a self-sustaining low-mood attractor. An earlier claim that the latent frame tracks an unshown worry item is withdrawn: the item was in the input, and the association is null after controls. Throughout we separate predictive claims from generative ones.

Contents

1 Introduction

2

2	Mental time travel and the depth regulator	3
2.1	Scope, and the role of optimism	3
2.2	Two logics of action and the identity pole	4
2.3	Retrieval constraints, interoception, and overgeneral memory	4
2.4	One depth regulator, two temporal faces	4
3	The temporal structure of free energy	5
3.1	Active inference and attention as precision	5
3.2	VFE as backward-looking evaluation	5
3.3	EFE as forward-looking evaluation	5
3.4	Constructionism and metacognition	6
4	The generative model	6
4.1	Framing actions as temporal-precision operators	6
4.2	The generative model in full, and where the frame enters	8
4.3	Extensions and the mood layer	8
5	Three-channel valence and the temporal taxonomy	8
5.1	The three channels	9
5.2	Composite valence and channel disagreement	9
5.3	A worked example	10
5.4	The VFE/EFE temporal taxonomy of emotion	10
6	Prior models are special cases	10
7	Temporal framing dynamics: self-regulation vs self-amplification	11
7.1	RECALL supports engagement, and its feedback reliance	12
7.2	Chronic stress as maladaptive temporal fixation	12
7.3	Sensitivity to the hand-set constants	14
8	Counterfactual emotions (a generative mechanism)	16
9	Empirical validation: head-to-head and extension	16
9.1	Subsumption: our affect layer recovers the happiness equation	18
9.2	Extension: the full model predicts affect dynamics, on two samples	18
9.2.1	Frame ablation: the frame does not carry the prediction	22
9.3	Latent frame and worry: a null after controls	22
9.4	Direct temporal-orientation test	23
9.5	What does not improve prediction	23
10	Clinical mapping: a two-level account of depressive affect	24
11	Discussion	24
11.1	What the integration buys	24
11.2	Metastability as health	25
11.3	Attention as precision, and mood/emotion timescales	25
11.4	Limitations	25

1 Introduction

Humans have a distinctive capacity for mental time travel. They simulate past, present, and future events to guide present decisions (Suddendorf and Corballis, 1997; Schacter and Addis, 2007). Moment-to-moment shifts of temporal orientation play out against more stable temporal

dispositions, letting people balance long horizons against the demands of the immediate. These shifts are inseparable from emotion: regret is directed at the past, hope at the future, surprise refocuses on the present. A central question is how past, present, and future are *instrumentally recruited* to guide behaviour toward preferred outcomes, and how emotions participate in that recruitment.

We approach this within predictive processing, on the view that emotional experience is an integral component of the inferential processes organising perception, learning, and action (Barrett, 2017; Seth and Friston, 2016). On this view affect is grounded in the agent’s success at minimising prediction error. It can be past-oriented, as the derivative of the trajectory of model fit (Joffly and Coricelli, 2013). Improving fit yields positive valence, deteriorating fit negative. It is also forward-looking. Expected free energy (EFE), minimised in policy selection (Parr et al., 2022), evaluates how well future observations will match preferences, and the “affective charge” of policy revision is itself a valence signal (Hesp et al., 2021). A present-oriented reward-prediction-error channel completes the picture (Pattisapu et al., 2025).

These accounts establish that affect has temporal structure, but they are *insufficient* in three respects. First, none unifies the variational/expected free-energy (VFE/EFE) distinction into a temporal taxonomy of emotion. Second, none models the single-agent mechanism by which an individual selects which temporal horizon of free energy dominates its affect, the mechanism Hinrichs et al. (2025) raised for dyads under the banner of “temporal aiming”, but left open within the agent. Third, each formalises only one temporal face of affect in isolation.

We close all three gaps. We (i) motivate the architecture from the psychology of mental time travel and cost-bounded simulation (§2); (ii) propose the VFE/EFE mapping as a unified theory of affective temporal direction and derive a three-channel valence decomposition (§3, §5); (iii) formalise counterfactual temporal framing as precision redistribution across horizons within a single factored generative model (§4); and (iv) show empirically that this integration is not merely conceptual. The integrated readout recovers prior single-channel models as special cases and improves on them where temporal structure exists (§9). Throughout, we separate claims that are *predictive* (the channel integration) from those that are *generative* (the counterfactual and hedonic-asymmetry mechanisms).

2 Mental time travel and the depth regulator

For much of everyday cognition, agents mentalise goals and the paths toward them, detaching from the present to simulate the future. Because remembering and future thinking recruit a shared constructive machinery (Schacter and Addis, 2007), these simulations are built out of memory. They must be precise enough to guide action, yet they are built with finite resources, a constraint that becomes acute as we juggle timescales from the immediate to the lifelong. Planning over a longer horizon *prima facie* requires more state-to-state transitions than a shorter one, but transitions are not free. Tree search over policies grows with branching factor and depth, so exhaustive planning is intractable for any bounded system, and each rollout step introduces approximation error that compounds along the trajectory.

2.1 Scope, and the role of optimism

The empirical literature on future-oriented cognition shows how humans handle this. At long horizons, little fine-grained arbitration takes place. Agents rely on broad, optimistic beliefs that preserve orientation without detailed simulation. Rather than constructing every future at fixed resolution, they “zoom” in and out across temporal scales; the number and resolution of intermediary steps vary with environmental complexity, novelty, uncertainty, and affective load. Near events are represented concretely, distant events abstractly and schematically (Trope and Liberman, 2010). Optimism is computationally sensible here. The optimal path to a distant goal

cannot be specified in advance, and filling in that far-off detail tends to provoke anxiety more than it yields mastery. On the broaden-and-build account (Fredrickson, 2001), positive affect widens the field of available thought and action and is less action-specific than negative affect, so future aiming is supported by beliefs that orient without specifying the future in full. Optimism here is the healthy, high-precision setting, not a universal default: under low positive-belief precision the same forward machinery turns anxious, yielding worry-type rather than hopeful prospection, a contingency we make explicit in the model (§4) and test directly (§9.4).

2.2 Two logics of action and the identity pole

Decision theory distinguishes two families of action selection (March, 1994). On the *logic of appropriateness*, the agent asks what kind of situation this is and what a person like them does here, recognising a situation and completing a pattern from a self-representation, rather than computing outcomes. On the *logic of consequences*, it lays out candidate action–outcome sequences and scores them against goals. The two differ in how deeply the plan is unrolled. Appropriateness conditions choice on slow, temporally extended beliefs about who one is and stays shallow; consequence-based control descends toward episode-level simulation. This maps a long-standing continuum between habitual (model-free) and goal-directed (model-based) control, but recasts it in *temporal* terms. A past-oriented mode leans on regularities slowly distilled from experience; a future-oriented mode is less tied to the current context.

2.3 Retrieval constraints, interoception, and overgeneral memory

How far the past can be recruited to serve the present is set by how autobiographical memory is actually constructed, and this is what a past-directed control operator in the model must respect. In the Self-Memory System (Conway and Pleydell-Pearce, 2000), autobiographical retrieval is a negotiation between *correspondence* (fidelity to experience-near sensory records) and *self-coherence* (pressure to keep memories consistent with the conceptual self); the working self gates construction as retrieval descends the autobiographical hierarchy. Whether the agent can reach a specific, self-congruent episode depends on its current affective and interoceptive state and its emotional granularity. Coarse, negatively valenced states narrow access to specific, positively toned material. The failure mode of the shallow setting is *overgeneral memory* (Williams et al., 2007). When coherence wins over correspondence, the cue is answered from the kind of person one takes oneself to be, not from an episode. This fixes what the past-directed operator of §4 must do. There the RECALL action pulls represented valence toward an identity-anchored target whose sign is set by the agent’s positive-belief precision (ρ_{pos} , the confidence of the slow mood belief that outcomes will be favourable; formalised in §4), so a coarse, negatively biased self answers a cue with overgeneral, negatively toned material, the failure mode just described. The abstract negotiation becomes one control variable in the model, how deeply retrieval descends the hierarchy.

2.4 One depth regulator, two temporal faces

These constraints force a control problem. Simulating the past or the future is unbounded and metabolically costly, so something must set how far any given memory or plan is unrolled. Left unregulated the system fails in one of two directions: it over-simulates, which is intractable and, as above, anxiogenic, or it under-simulates, which yields overgeneral memory looking back and absent foresight looking forward. We call the quantity that sets this depth the *depth regulator*, and it is what the rest of the model controls. It is one regulator rather than two because the same knob, how far to descend into constructed episodes, operates in both temporal directions. It weighs identity fit against computational cost, and it has two temporal faces. On the past side, the agent arbitrates between self-coherence and correspondence; its healthy shallow setting is a

positively biased working self. On the future side, the healthy shallow setting is orientation without detailed simulation; a plan is committed to only once computed and broken into steps, with depth bounded by cost. The two directions of failure recover the classical split (Nolen-Hoeksema et al., 2008): *rumination* (past-directed, self-focused) when a habitual, identity-congruent bias overrides the escape that consequence-based control would favour; *worry* (future-directed, anticipatory) when a bias toward reward overrides the cost that limits forward simulation (Ehring and Watkins, 2008). This double bind is the computational core of the chronic-stress pathology of §7. The remainder of the paper gives the depth regulator a formal treatment in the language of free energy. In the model that follows, its continuous depth is discretized into a three-valued temporal-frame factor and six framing actions that redistribute precision over it. The discretization is a simplification, flagged again in the limitations, and does not assert that regulated depth is intrinsically discrete.

3 The temporal structure of free energy

3.1 Active inference and attention as precision

Active inference holds that agents encode a generative model relating hidden states to observations and minimise free energy across perception, learning, and action. Prediction errors, discrepancies between expected and sensed observations, drive belief updates, weighted by *precision*, the gain that determines which errors shape belief. Precision-weighting is attention (Feldman and Friston, 2010), and agential control over it requires a metacognitive architecture in which higher-order states modulate which lower-level processes receive precision (Sandved-Smith et al., 2021). It is this second-order control of precision over temporal horizons that we formalise as temporal framing.

3.2 VFE as backward-looking evaluation

For a generative model $p(o, s)$ with approximate posterior $q(s)$,

$$F = \underbrace{-\mathbb{E}_{q(s)}[\ln p(o | s)]}_{\text{accuracy}} + \underbrace{D_{\text{KL}}[q(s) \| p(s)]}_{\text{complexity}}, \quad (1)$$

which scores how well the *current* model fits *past* observations. Joffily and Coricelli (2013) showed that its temporal derivative is a valence signal: decreasing F (improving fit) is positive valence, increasing F negative. The second derivative differentiates its dynamics: accelerating improvement, decelerating improvement toward a plateau, and their negatives give a family of affective states from a single backward-looking quantity. These are dynamics of *past* model fit and are distinct from the forward-looking hope and fear of §5, which evaluate *future* observations, even where ordinary language reuses the words.

3.3 EFE as forward-looking evaluation

For a policy π producing predicted observations,

$$G(\pi) = \sum_m \left[\underbrace{D_{\text{KL}}[q(o_m | \pi) \| p(o_m | C_m)]}_{\text{risk}_m} + \underbrace{\mathbb{E}_{q(s)}[H[p(o_m | s)]]}_{\text{ambiguity}_m} \right], \quad (2)$$

where C_m encodes preferences. EFE asks not “did I predict well?” but “will I predict well?”. Its risk term is expected value relative to preference; this is the quantity a forward affective channel reads out, and the quantity that supplies *anticipated value* when the channels are placed on a decision task-model.

3.4 Constructionism and metacognition

By *agent* we mean the organism modelled as an active-inference system, not a homunculus inside it, and we take no stance on phenomenal consciousness; the distinction we rely on is between levels of control, not between conscious and unconscious experience. Running automatically, without second-order intervention, the base perception–action loop reproduces a flow-like present engagement with the world (Rietveld and Kiverstein, 2014). Made multiscale, it affords a constructionist account of emotion (Barrett, 2017). Emotions are concepts moulded onto core processes of allostatic regulation and interoceptive monitoring (Seth and Friston, 2016; Stephan et al., 2016). Metacognitive control of attention, a covert mental action that modulates which processes receive precision, requires a second-order layer (Sandved-Smith et al., 2021), higher-order states setting lower-level precision, and it is here that counterfactual temporal framing lives. Remembering and imagining are a single offline generative operation, the same model run over past or future states instead of present ones (Schacter and Addis, 2007).

4 The generative model

We adopt a factored POMDP with hidden state $s = (v, e, f)$: **valence** $v \in \{0, \dots, K-1\}$ at granularity K ; **energy** $e \in \{0, \dots, M-1\}$, an interoceptive resource estimate; and **temporal frame** $f \in \{\text{PAST}, \text{PRESENT}, \text{FUTURE}\}$, the active counterfactual orientation. Three observation modalities, external feedback o_{ext} , interoception o_{int} , and felt valence o_{val} , couple to these factors. The frame f is a hidden state the agent *infers*, not a label it assigns. The posterior $q(f | o)$, marginalised over v and e , is a direct readout of believed temporal orientation.

Two quantities in this paper carry the name valence, and separating them removes an apparent circularity. The hidden state v is *represented* valence, the agent’s coarse posterior belief about its own affective condition, inferred jointly from external feedback, interoception, and a felt-valence signal, and used to gate retrieval and to report. The three *valence channels* of §3 are not states but signals read off the dynamics of inference itself: the backward channel is the rate of change of variational free energy ($-dF/dt$), the present channel a reward prediction error, the forward channel the affective charge of policy revision. These channels generate the felt-valence observation o_{val} , and the state v is what the agent infers from it. Valence is therefore neither simply given nor only a derivative of free energy. The dynamic signals drive the felt observation, and v is the belief the agent forms about its affective state from that observation together with its other evidence. This split is the sense in which affect is a readout, sitting on top of the task model rather than replacing it. Figure 1 shows the factor graph. To read it, note that the agent receives three observations (the coloured outcome squares: external feedback, interoception, and felt valence), infers the three hidden factors that generate them through a likelihood $\mathbf{A}$ (valence, energy, temporal frame), and moves each factor through a controllable transition $\mathbf{B}$ by selecting a policy, scored by expected free energy G . Red arrows are the message passing that carries beliefs between them. It is one task-specific model, and the three affective channels are three readings taken from this single loop rather than three separate models.

4.1 Framing actions as temporal-precision operators

Each action defines a transition $\mathbf{B}_a = \mathbf{B}_v^{(a)} \otimes \mathbf{B}_e^{(a)} \otimes \mathbf{B}_f^{(a)}$, whose frame block $\mathbf{B}_f^{(a)}$ is a *temporal-precision operator* that redistributes probability across horizons. Six actions realise the depth regulator:

RECALL assigns 70% self-transition to PAST; $\mathbf{B}_v^{\text{RECALL}}$ pulls valence toward a *bidirectional* target $v_{\text{recall}} = 0.2 + 0.6\alpha$ with $\alpha = \sigma(\rho_{\text{pos}} - \theta)$, so high positive-belief precision ρ_{pos} recalls a positive past (narrative stabilisation) and low ρ_{pos} a negative one (rumination). It is the computational realisation of *shallow*, identity-anchored retrieval (§2.4).

factor graph | magenta: outcomes (A) cyan: hidden states (B) blue: controllable puce: policies (G)
red arrows: 1 ascending likelihood, 2 forward, 3 backward messages

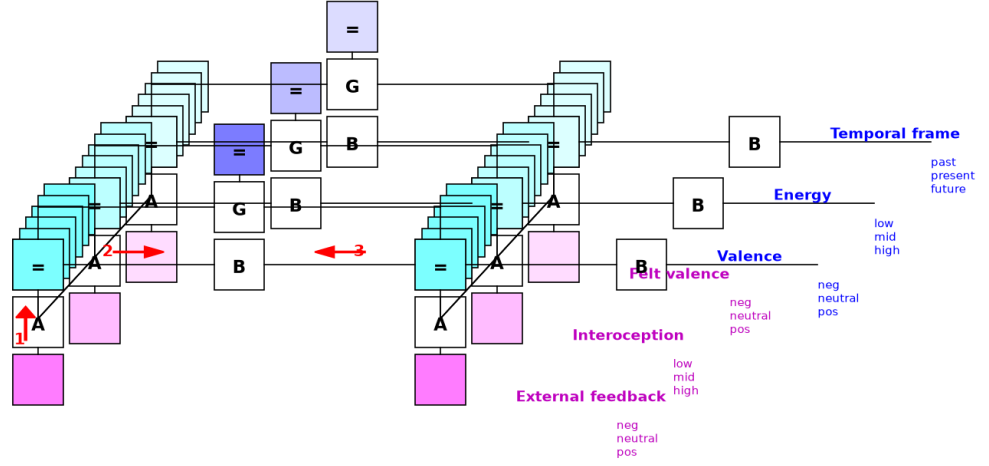


Figure 1: **Factor graph of the generative model** (SPM convention: magenta = outcome modalities with likelihood A ; cyan = hidden-state factors with transition B ; controllable factors are stacks with policy and expected-free-energy G nodes; red arrows = message passing). Three controllable factors (valence, energy, temporal frame) and three modalities integrate the backward, present, and forward affective channels in one model.

ENGAGE assigns 75% to PRESENT, maintaining precision on current sensory input through perception–action coupling.

FUTURATE assigns 90% to FUTURE (narrative inertia); B_v^{FUTURATE} pulls valence toward a *precision-gated* target $v_{\text{fut}} = 0.2 + 0.6\alpha$, symmetric with RECALL, so high positive-belief precision yields hopeful prospection and low precision yields anxious, worry-type prospection rather than fixed optimism. Its energy cost brakes it into short reactive planning bursts.

FEEL releases frame commitment (50–60% present) and models active interoceptive processing, reducing accumulated prediction error through body-state monitoring (Stephan et al., 2016).

DISSOCIATE distributes probability roughly uniformly across frames, modelling *temporal unmooring* (Vannikov-Lugassi and Soffer-Dudek, 2018).

ABSTRACT assigns ≈ 60 –80% to FUTURE, an effortful, ungrounded forward route that bypasses interoception and episodic recall.

B_{frame} matrices are additionally learned online via Dirichlet updates, so agents acquire personalised temporal-attention operators. Here σ is the logistic function and $\theta = 2$ the precision threshold; the exact frame- and valence-transition entries, the channel temperatures τ_{model} , τ_{reward} , τ_{action} , and the mood-layer constants are given in the supplementary implementation (`generative_model.py`, `agent.py`). The self-transition entries above (0.70, 0.75, 0.90, 0.80) and the valence pull weights are set by hand. §7.3 sweeps them and reports which qualitative results survive.

4.2 The generative model in full, and where the frame enters

The likelihoods factorise as $p(o_{\text{ext}} | v, e)$, $p(o_{\text{int}} | e)$ and $p(o_{\text{val}} | v)$. None depends on f : the frame changes no predicted observation directly, which is why it cannot be recovered from a single affect reading (§9.4). Transitions are $\mathbf{B}_a = \mathbf{B}_v^{(a)} \otimes \mathbf{B}_e^{(a)} \otimes \mathbf{B}_f^{(a)}$, preferences C_m are over observations, and the prior $D(v, e, f) \propto e^{\rho_{\text{pos}}(v_n - 0.5)} e^{-3(e_n - 0.7)^2} [0.2, 0.6, 0.2]_f$ encodes positive self-belief precision ρ_{pos} (written `pi_pos` in the code and figure axes). Beliefs update as $q(s_t) \propto \mathbf{B}_{a_{t-1}} q(s_{t-1}) \prod_m p(o_{m,t} | s_t)$.

The frame enters through the posterior marginal $q(f)$, which sets three *horizon precisions*

$$w_c = 1 + g(3q(f=c) - 1), \quad c \in \{\text{PAST}, \text{PRESENT}, \text{FUTURE}\}, \quad (3)$$

which sum to 3 for every $q(f)$ and reduce to $(1, 1, 1)$ at gain $g = 0$. They weight the affective channels in the readout, $V = \tanh(w_{\text{PAST}} v_{\text{model}} + w_{\text{PRESENT}} v_{\text{reward}} + w_{\text{FUTURE}} v_{\text{action}})$, and the horizon terms of the expected free energy that scores each action,

$$G(a) = w_{\text{PRESENT}} G_1(a) + w_{\text{FUTURE}} \delta \sum_{a'} \pi'(a' | a) G_1(a' | a) + w_{\text{PAST}} D_{\text{KL}}[q(s_{t+1} | a) \| D], \quad (4)$$

with G_1 the one-step risk plus ambiguity of §3, $\delta = 0.75$ the discount on a one-step counterfactual rollout π' (the roads not taken), and the last term the divergence of the predicted next state from the identity prior, the retrospective cost a past-oriented agent pays for leaving its established self-narrative. Policies follow $\ln \pi(a) = -\gamma G(a) + E_a$. So a future-dominant posterior deepens prospection and amplifies the forward channel, a present-dominant one weights immediate risk and reward prediction error, and a past-dominant one weights narrative consistency and the backward channel. Framing actions move $q(f)$ through $\mathbf{B}_f^{(a)}$, and $q(f)$ in turn changes which horizon the next action is scored on: this is the feedback loop between inferred frame and policy evaluation. The gain is $g = 1$ throughout; $g = 0$ is the ablation in which the frame is inferred but gates nothing, which was the model in an earlier version of this paper, where the frame was an action-conditioned label without a causal role (a point we owe to the AAAI-27 reviewers).

At the slow timescale, the mood layer holds a categorical posterior $q(m)$ over discretised values of ρ_{pos} , updated every $T_{\text{mood}} = 50$ steps from the mean believed valence $\bar{v}$ over the window, $q(m) \propto \mathcal{N}(\bar{v}; 0.5 + 0.02(m - \theta), 0.06^2) [\mathbf{B}_{\text{mood}} q(m)]$, with $\rho_{\text{pos}} \leftarrow \mathbb{E}_{q(m)}[m]$ then rebuilding the valence transitions. Neutral valence is anchored to the rumination knee θ .

4.3 Extensions and the mood layer

Two extensions capture affective pathology. *Asymmetric hedonic sensitivity* replaces the scalar reward sensitivity with a pair $(c_{\text{pos}}, c_{\text{neg}})$ scaling positive and negative preference entries separately, capturing dissociable reward and punishment processing (Eshel and Roiser, 2010; Pizzagalli, 2014). *Habit priors* add an additive log-prior $\mathbf{E}$ over policies (Da Costa et al., 2020), so $\ln \pi(a) = -\gamma G(a) + E_a$, modelling EFE-suboptimal habitual tendencies such as ruminative fixation (Watkins, 2008). A hierarchical *mood layer* (M5) infers ρ_{pos} over slow timescales as a categorical POMDP driven by accumulated believed-valence relative to neutral (Eldar et al., 2016), separating within-trial emotion from cross-trial mood (Hesp et al., 2021) and producing a self-reinforcing low-mood attractor under vulnerability and stress from Bayesian inference alone (§7).

Table 1 maps each choice to its source; full detail is in the supplementary design-justification document.

5 Three-channel valence and the temporal taxonomy

The distribution of attention across horizons is a second-order inference, and the three channels are read at successive stages of a single agent–environment cycle. Updating beliefs from a new

Table 1: **Design justification (excerpt)**. Predictive (P) and generative (G) status flagged where the distinction applies.

Choice	Rationale	Source
Attention = precision (P)	framing is precision redistribution over horizons	Feldman and Friston (2010)
Backward channel (P)	$-dF/dt$ valence	Joffily and Coricelli (2013)
Present channel (P)	reward prediction error	Pattisapu et al. (2025)
Forward channel (P)	affective charge of policy revision (EFE)	Hesp et al. (2021)
Temporal-frame factor	mental time travel; shared memory/imagination machinery	Schacter and Addis (2007); Suddendorf and Corballis (1997)
Energy/interoception	allostatic body-budget	Seth and Friston (2016); Stephan et al. (2016)
RECALL (past)	self-memory system; overgeneral memory on failure	Williams et al. (2007)
Habit priors (E-vector)	habitual, EFE-suboptimal tendencies	Da Costa et al. (2020); Watkins (2008)
Hedonic asymmetry (G)	reward/punishment sensitivity dissociate	Eshel and Roiser (2010); Pizzagalli (2014)
Counterfactual emotions (G)	regret/relief from obtained vs foregone	Hesp et al. (2020)
Mood layer (M5)	mood/emotion timescale separation	Hesp et al. (2021); Eldar et al. (2016)
Effort/energy cost	effortful deliberation carries a metabolic cost	Treadway et al. (2009)

observation exposes the backward channel, model valence. Comparing obtained against expected reward exposes the present channel, reward valence. Revising the policy exposes the forward channel, action valence.

5.1 The three channels

Model valence (backward) tracks $v_{\text{model}} = \tanh(-\Delta F/\tau_{\text{model}})$ (Joffily and Coricelli, 2013), modulating the model learning rate. *Reward valence* (present) is $v_{\text{reward}} = \tanh((U - \mathbb{E}[U])/\tau_{\text{reward}})$ over hedonic modalities (Pattisapu et al., 2025); interoceptive prediction errors drive allostatic regulation but are excluded from felt valence. *Action valence* (forward) is $v_{\text{action}} = \tanh(\text{AC}/\tau_{\text{action}})$ with $\text{AC} = (\pi_{t-1} - \pi_t) \cdot \mathbf{G}$ (Hesp et al., 2021), and feeds back on policy precision. This closes the emotion–action loop: policy revision changes expected free energy, action changes what is observed, and the resulting change in variational free energy feeds back through v_{action} onto the precision that governs the next policy.

5.2 Composite valence and channel disagreement

The channels combine as $V = \tanh(v_{\text{model}} + v_{\text{reward}} + v_{\text{action}})$. Agreement yields unambiguous satisfaction or distress; disagreement is informative. *Productive failure* ($v_{\text{model}} < 0$, $v_{\text{action}} > 0$): the outcome was bad but a better plan was found. *Unsettling success* ($v_{\text{reward}} > 0$, $v_{\text{action}} < 0$): good fortune that may not last. *Dawning anxiety* ($v_{\text{model}} \approx 0$, $v_{\text{reward}} \approx 0$, $v_{\text{action}} < 0$): nothing has happened, but plans are deteriorating.

5.3 A worked example

Take one trial of the gamble task-model of §9. The agent gambles and wins, and the win is larger than it expected. Read the channels in turn. The present channel compares obtained reward with its expectation; the surplus makes $v_{\text{reward}} > 0$. The backward channel registers that the win improved model fit, so F fell and $v_{\text{model}} > 0$. The forward channel registers that policy precision has shifted toward gambling and that this lowered expected free energy, so its affective charge $v_{\text{action}} > 0$. All three agree; the composite $V = \tanh(v_{\text{model}} + v_{\text{reward}} + v_{\text{action}})$ is strongly positive, and the state is unambiguous satisfaction. Now change one thing: let the gamble lose, but let the loss reveal that the safe option was the better policy. The present and backward channels turn negative (the outcome was bad and fit worsened) while the forward channel stays positive (a better plan was found). The channels disagree, and the composite reports the muted, forward-leaning affect labelled productive failure above. These three readings, taken at three stages of one inference cycle, are all the taxonomy requires.

5.4 The VFE/EFE temporal taxonomy of emotion

Emotions inherit their temporal direction from whichever quantity generates them, yielding a taxonomy we offer as an organising scheme rather than an essentialist classification; its partitioning within an agent depends on developmental and sociocultural history (Barrett, 2017). *Backward* ($-dF/dt$, the rate of change of VFE): regret, guilt, shame, pride, gratitude. *Present* (RPE): joy, anger, disgust, surprise. *Forward* (EFE): hope, fear, anxiety, anticipation. *Transitional* (VFE vs prior EFE): relief, disappointment, satisfaction. This grounds the temporal axis of appraisal theory (Smith and Ellsworth, 1985; Ortony et al., 1988) in a formal distinction rather than folk labels.

The taxonomy is more useful as a set of discriminating predictions than as a list of labels, so we state the ones the model commits to. *Relief versus disappointment* are transitional: both require a prior forward expectation and a present outcome, so both should vanish when the forward channel is silenced (a task with no policy revision), whereas joy and anger, present-channel emotions, should not. *Regret versus disappointment* separate on the counterfactual: regret requires a foregone alternative that was better and should appear only under complete feedback, disappointment only needs an outcome below expectation and should appear under partial feedback too (§8 tests the regret half of this on choice data). *Anxiety versus fear* separate on the frame: with the same forward channel value, a future-dominant posterior $q(f)$ should amplify the affective charge (Eq. 3) and lengthen its persistence, so anxiety is fear under future fixation, which is the mechanism behind the stressed phenotype of §7. *Pride versus joy* separate on the backward channel: pride requires that the outcome improved model fit relative to a self-relevant prior, joy does not. Each of these is a within-person contrast a suitably designed task can test, and each names the channel or frame manipulation that should abolish it.

6 Prior models are special cases

Figure 2 shows the architecture as a stack of layers and where the prior models sit inside it. A task-specific generative model sits at the base: states, observations, policies, and expected free energy for whatever task the agent faces (the gamble task-model of §9, an experience-sampling affect model, any decision problem). On top of it sits the affective readout layer, the three channels, whose composite feeds a represented valence state and a slow mood layer, with the temporal-frame factor modulating precision across horizons. Each prior model is one layer of this stack, and our contribution is the integration, not any single channel.

The three predecessor accounts are *reduced sub-graphs* of our model (Figure 3, schematic): Joffily and Coricelli (2013) is a perception-only VFE model (one factor, no policy, backward channel only); Pattisapu et al. (2025) is a single-factor POMDP with policy selection (present

channel, no temporal frame, no backward derivative); Hesp et al. (2021) is a deep hierarchical model carrying the forward affective-charge channel and a slow mood layer, but without a reward decomposition or a temporal-frame factor. Each predecessor contributes at most one of our affective channels, and we make the debt explicit: the mood layer that carries our diathesis-stress dynamics is adopted from Hesp et al. (2021) (with its evidence changed from free energy to valence, §7). The contribution is thus an integration, a task-specific model, and an inferred temporal frame that none of the three has, not a claim to have out-built any one of them on its own terrain; §9 shows the channel decomposition precisely.

Affect as a readout layer over a task-specific model: the whole architecture, and where prior models sit

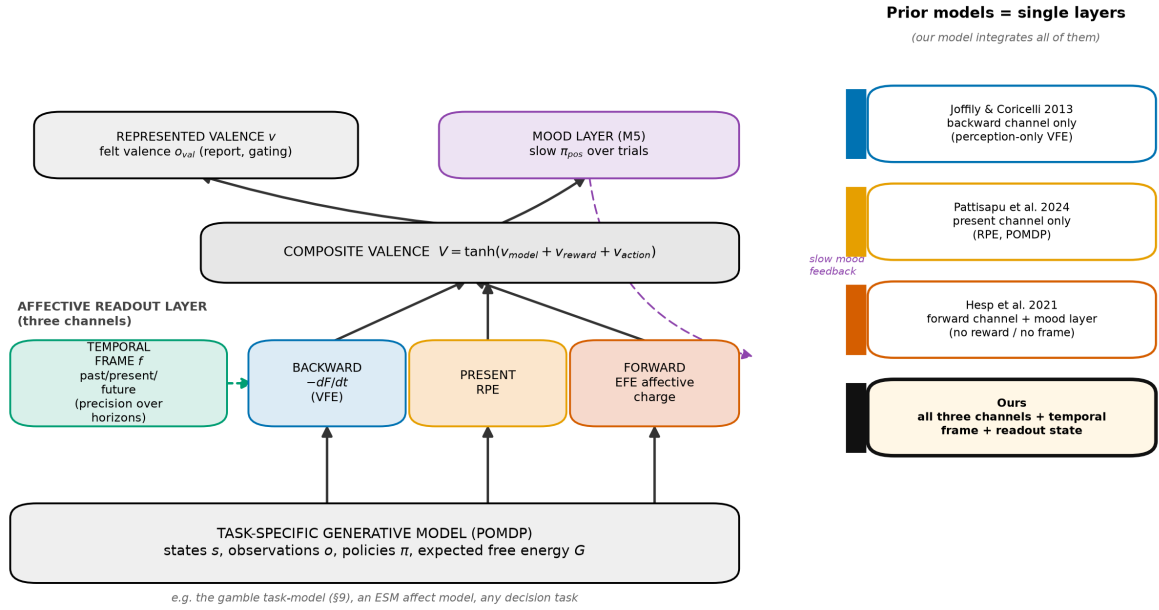


Figure 2: **The layered architecture, and where prior models sit.** Affect is a readout layer over a task-specific generative model (base). The three channels (backward, present, forward) read off the dynamics of inference; their composite feeds a represented valence state and a slow mood layer, with the temporal-frame factor modulating precision across horizons. Each prior model corresponds to a single layer: Joffily & Coricelli to the backward channel, Pattisapu et al. to the present channel, Hesp et al. to the forward channel plus mood layer. Our model integrates all of them and adds the temporal-frame factor and the readout state.

7 Temporal framing dynamics: self-regulation vs self-amplification

Adaptive flexibility is the capacity to move between framing modes as the situation demands. The loop admits two regimes. Under *self-regulation* it is homeostatic. High VFE triggers corrective action, and positive v_{action} sharpens precision on the corrective trajectory. Under *self-amplification* it is pathological. Framing locks into a single mode; precision on that mode’s errors intensifies, and the agent cannot update away. We illustrate the dynamics across three parameter phenotypes used in Figure 4: a *healthy* agent (high ρ_{pos} , intact interoception), a *depressive* one (low ρ_{pos} , blunted reward sensitivity), and a *manic* one (low interoceptive precision and disinhibited forward framing).

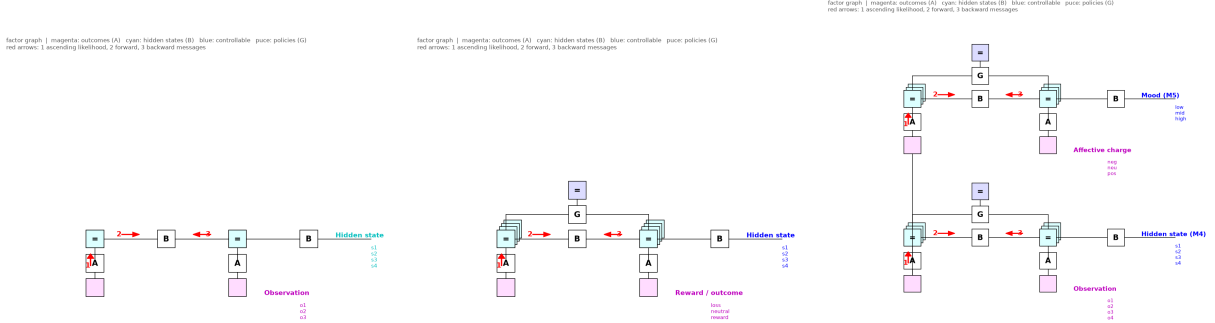


Figure 3: **Predecessor models as reduced factor graphs.** Joffily & Coricelli (backward, perception-only); Pattisapu et al. (present, single-factor POMDP); Hesp et al. (forward, deep hierarchy). Each captures one channel; none has the temporal-frame factor of Figure 1.

7.1 RECALL supports engagement, and its feedback reliance

Past-oriented framing is preparatory: RECALL stabilises beliefs by pulling toward the prior, lowering VFE and making ENGAGE more competitive, so a RECALL-to-ENGAGE pipeline emerges in healthy agents (Figure 4d,e). This capacity is constrained by feedback. When ρ_{pos} is low, RECALL fails at two coupled levels: the mechanism itself is impaired ($\alpha \approx 0.14$, barely pulling toward the prior) and its deployment is suppressed (it becomes EFE-uncompetitive). Figure 5 contrasts a healthy and a recall-impaired agent differing only in ρ_{pos} : RECALL utilisation collapses from 44% to 0%, with the freed policy mass going almost entirely to FEEL (18% to 68%), the interoceptive null action. This is the computational form of overgeneral memory (Williams et al., 2007): with identity-anchored retrieval unavailable, the agent falls back on monitoring its body state rather than on constructive temporal processing.

7.2 Chronic stress as maladaptive temporal fixation

Chronic stress realises the double bind of §2.4. Affect turns negative but construal stays coarse, so neither pathway to policy selection delivers. A stressed profile (very high volatility, amplified reward sensitivity, stress-impaired interoception, reduced ρ_{pos}) shows future fixation: FUTURE frame belief 0.44 against 0.20 for the healthy agent, PRESENT depleted (0.40 vs 0.51), ABSTRACT, the low-cost prospection operator, taken on 40% of steps against 3%, and negative reward valence (-0.15 vs $+0.04$) from *frustrated forward projection*: abstract prospection commits to above-neutral predictions that a volatile world falsifies (Figure 6). The PAST frame stays the least probable throughout (0.16); the stressed agent does not transition into past-oriented rumination within the episode. The direction of these effects holds across seeds and across the constant sweep of §7.3. The stressed profile differs from the healthy one on four parameters at once, so this is a phenotype, not a one-factor manipulation; the sweep in §7.3 additionally varies each of the four from the healthy profile alone.

At the slow mood timescale, the M5 layer accumulates believed-valence evidence, so persistently below-neutral affect drives positive-belief precision ρ_{pos} below the rumination knee, where the RECALL target turns negative. The transition from future-directed worry to past-directed rumination is therefore a prediction of the mood layer over months, not something the within-episode simulation of Figure 6 shows; we present it as a model-generated hypothesis awaiting longitudinal data. What the mood layer does produce in simulation is a *diathesis-stress* pattern (Figure 7). Crossing vulnerability (low ρ_{pos} , blunted reward sensitivity, weak interoception) against chronic stress (high volatility) in a 2×2 design, only the conjunction collapses. Across eight seeds a healthy agent stays high even under chronic stress (final $\rho_{\text{pos}} = 7.4 \pm 0.1$, depressed in 0/8), vulnerability alone recovers (5.1 ± 0.6 , 0/8), and only the vulnerable agent under stress falls into a self-sustaining depressive attractor (1.3 ± 0.3 , 8/8 below the knee). Neither factor

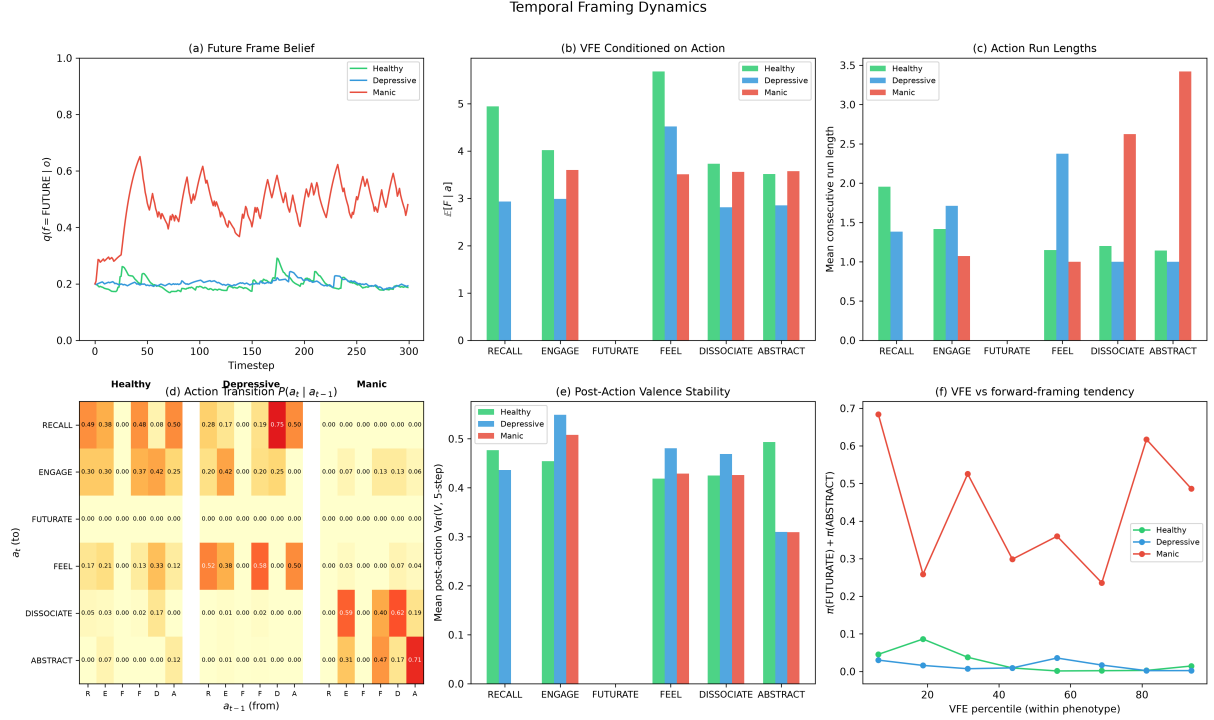


Figure 4: **Temporal framing dynamics.** (a) Future-frame belief over time per profile. (b) Mean VFE conditioned on the selected action. (c) Action run lengths (ABSTRACT and DISSOCIATE longest in the manic profile). (d) Action transition matrices: the RECALL-to-ENGAGE pipeline in healthy agents. (e) Post-action valence stability: RECALL lowest. (f) Forward-framing tendency ($\pi(\text{FUTURATE}) + \pi(\text{ABSTRACT})$, the two future-pulling operators, of which ABSTRACT carries most of the mass) against within-profile VFE percentile. FUTURATE itself is a rare, high-cost reactive-planning burst; ABSTRACT is the low-cost operator that does most forward framing.

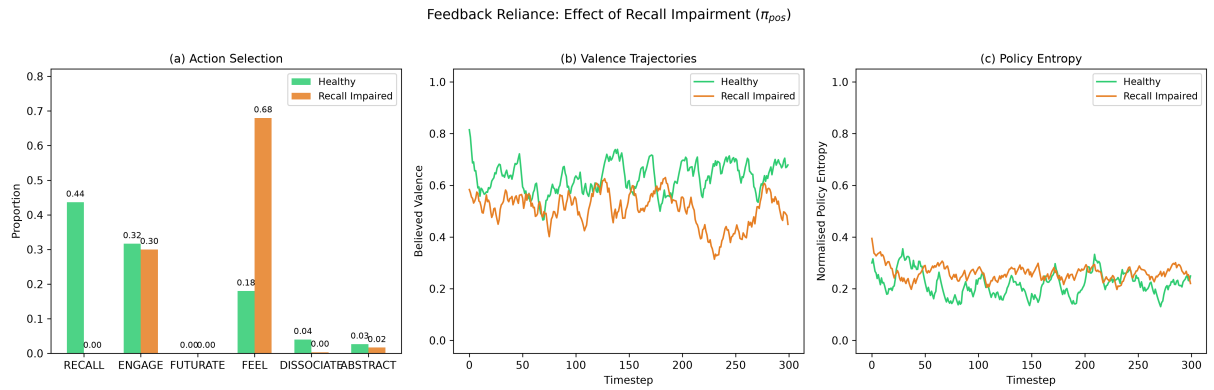


Figure 5: **Feedback reliance under recall impairment.** Profiles share all parameters except ρ_{pos} (5.0 vs 0.2); frame-gated model. (a) RECALL drops from 44% to 0%, with the mass moving to FEEL. (b) Valence trajectories lower under impairment. (c) Normalised policy entropy (smoothed): the impaired agent runs slightly higher for most of the episode, so its policy is marginally less decisive; the difference is small (§7.3 reports it across seeds), and in the frame-inert version of the model its sign was reversed, a caption error in an earlier version of this paper that the AAAI-27 reviewers caught.

alone suffices, matching the diathesis-stress account of depression (Monroe and Simons, 1991) and the resilience of most people to stress.

The mechanism instantiates vulnerability rather than fitting it, but its central prediction is testable in real data, and it holds. The mechanism entails that a vulnerability trait should both lower baseline mood and *amplify* affective reactivity to stress, a vulnerability $\times$ stress interaction rather than either main effect. In the Geschwind ESM, baseline neuroticism, which the affect-dynamics fitting never uses, does exactly this: it lowers baseline mood ($r = -0.53$, $n = 128$) and steepens each person’s affective response to daily events (between-person $r = 0.31$; pooled event $\times$ neuroticism interaction $t = 4.2$ over 11,315 beeps), reproducing the established neuroticism $\times$ stress reactivity effect (Geschwind et al., 2011). What this 20-day, already-remitted sample cannot show, and the mechanism additionally predicts, is the months-scale transition into *sustained* low mood from the same conjunction, preceded by a persistent below-neutral shift in momentary valence rather than in prediction error, a test that awaits longer longitudinal data.

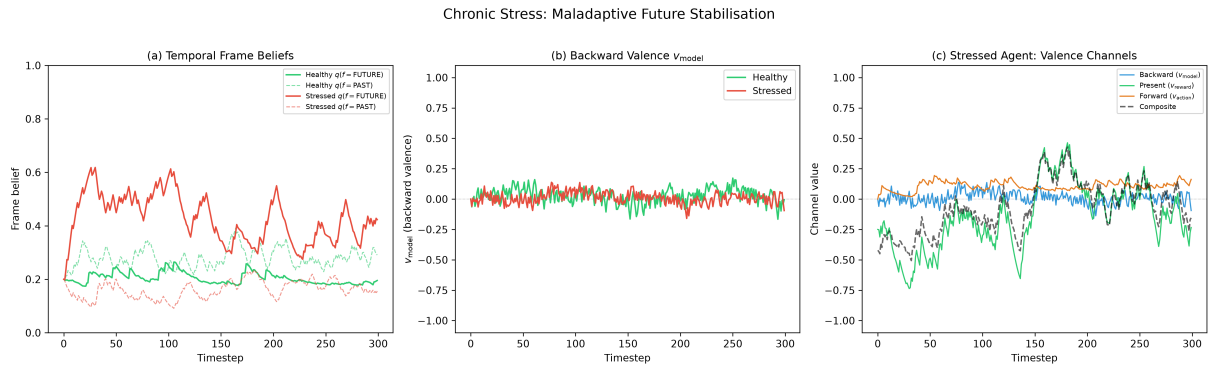


Figure 6: **Chronic stress as maladaptive temporal fixation.** Stressed vs healthy, frame-gated model. (a) Frame beliefs: the stressed agent’s FUTURE belief (solid red) dominates and its PAST belief (dashed) stays lowest; there is no transition to past framing within the episode. (b) Backward valence, similar in both agents. (c) Channel decomposition for the stressed agent: the present channel (reward prediction error) is negative for most of the run and drives the composite down; the forward channel stays mildly positive (policy revision under volatility).

7.3 Sensitivity to the hand-set constants

The frame self-transition entries and the valence pull weights of the framing operators are set by hand, so we swept them. Figure 8 varies the target-frame stickiness s over $\{0.5, \dots, 0.95\}$ (replacing the four defaults at once) and the pull-weight scale k over $\{0.6, \dots, 1.4\}$, four seeds per cell, and separately varies the gating gain g from 0 (frame-inert) to 1. Three results are scored: the RECALL collapse under impairment, future fixation under stress, and the diathesis-stress conjunction. The RECALL collapse is positive in every cell of the 30-cell grid (range 0.21 to 0.39), and future fixation is positive in every cell (range 0.00 to 0.32), growing with both stickiness and pull weight and vanishing only when the pull weights are cut to 0.6 of their default, where prospection no longer moves valence. The diathesis-stress result holds in all four seeds in 77% of cells and in a majority of seeds in 87%; the failures are all at $k = 1.4$, where the amplified pulls let the vulnerable agent under stress climb back above the knee in some seeds. Against the gain, all three results hold at every g ; the RECALL collapse is larger with gating (0.37 to 0.41 against 0.26 at $g = 0$) and future fixation smaller (0.11 to 0.25 against 0.27), because a past-weighted retrospective term makes RECALL more attractive to the healthy agent and a present-weighted risk term brakes ABSTRACT in the stressed one. The one quantity whose sign depends on the constants is the policy-entropy difference of Figure 5c: it is small and positive across most of the grid but negative at $k = 1.4$ and at $s = 0.95$, and negative at $g = 0$. We therefore do not build

Diathesis-Stress: Depression Emerges from Vulnerability $\times$ Chronic Stress (Hierarchical M5 Mood Layer)

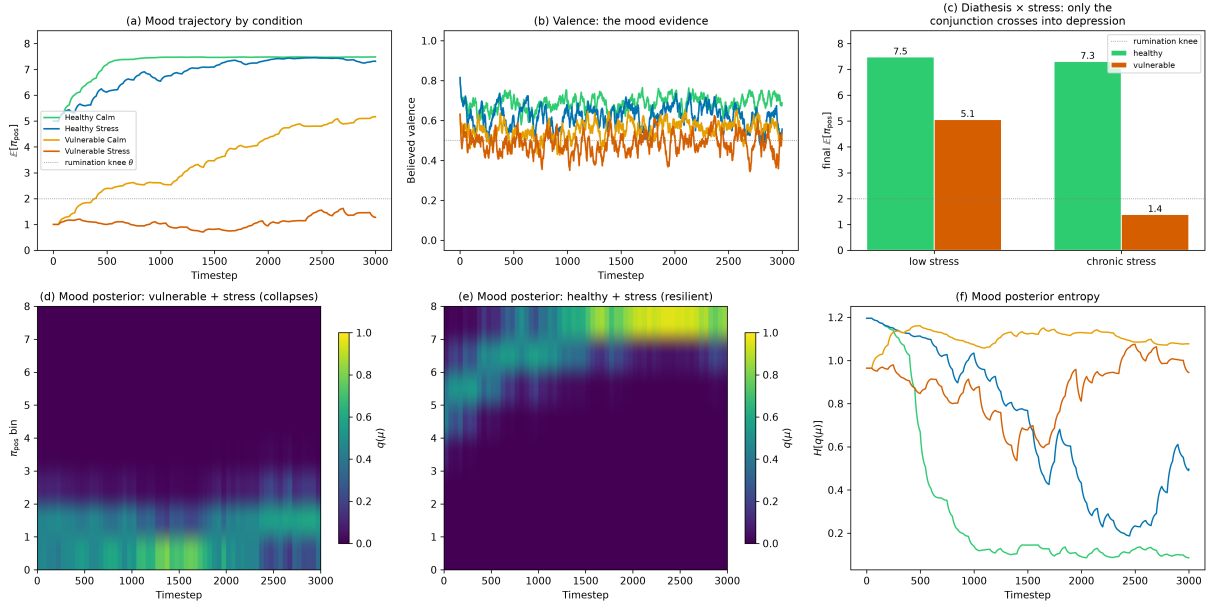


Figure 7: **Diathesis-stress in the M5 mood layer.** A 2×2 design crosses vulnerability (low ρ_{pos} , blunted reward sensitivity, weak interoception) with chronic stress (high volatility). (a) Mood ($\mathbb{E}[\rho_{\text{pos}}]$) trajectories; (b) believed valence, the mood evidence, relative to neutral (dotted). (c) Final mood by condition: only vulnerability $\times$ stress crosses the rumination knee ($\theta = 2$). (d) The vulnerable-under-stress mood posterior collapses to the low bin; (e) the healthy-under-stress posterior stays high (resilience). (f) Mood-posterior entropy. Holds across seeds (vulnerable+stress depressed 8/8; others 0/8). $T = 3000$.

on it.

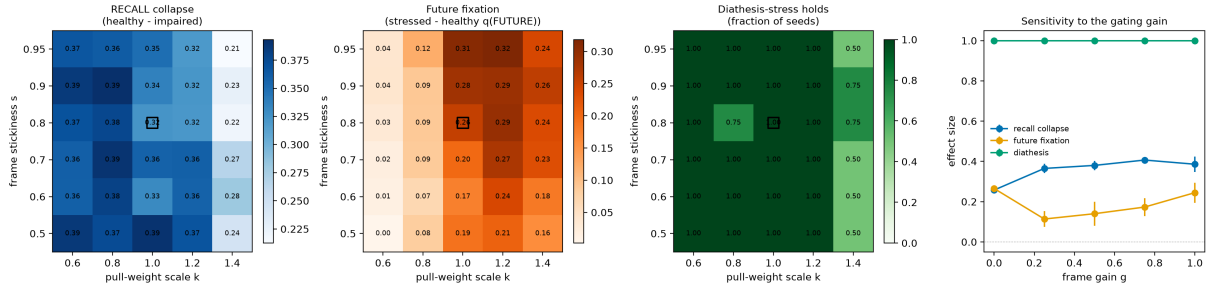


Figure 8: **Sensitivity sweep.** Left three panels: frame stickiness s (the self-transition entry of each framing action’s target frame, replacing 0.70/0.75/0.90/0.80) against the valence pull-weight scale k (FUTURATE $0.5k$, ABSTRACT $0.65k$), four seeds per cell, black square at the paper’s defaults. Cells show the RECALL collapse (healthy minus impaired RECALL proportion), future fixation (stressed minus healthy $q(\text{FUTURE})$), and the fraction of seeds in which the diathesis-stress result holds (vulnerable-under-stress below the knee, healthy-under-stress above). Right: the same three effects against the frame gain g , with $g = 0$ the frame-inert model. Script `sensitivity_bframe.py`.

8 Counterfactual emotions (a generative mechanism)

Regret and relief require comparing what happened with what *would* have happened. Adapting the counterfactual free-energy machinery of Hesp et al. (2020), we write $\text{Regret} = F_{\text{actual}} - \min_{\pi'} F_{\text{counterfactual}}(\pi')$. Computing $F_{\text{counterfactual}}$ requires running the generative model offline, exactly what temporal framing does. Its behavioural signature is switching: in complete-feedback choice tasks the outcomes are binary, so “regret” (foregone better than obtained) is confounded with “obtained a loss”, and the honest contrast is at fixed obtained outcome. After a loss, participants switch on the next visit to the same state more often when the foregone option was better than when it was equally bad (0.46 vs 0.33, paired $t = 10.4$ over 143 participants; Figure 9); the same contrast replicates on a second complete-feedback dataset (Palminteri et al., 2017) (0.43 vs 0.16, $t = 7.3$, $n = 20$).

We then asked what fitted models generate on the same trials. A factual Q-learner (chosen option only) produces part of the contrast without any counterfactual input, +0.05 against the human +0.13, because states in which the other option is good are states in which its learned value is higher. Adding the regret signal as a policy bias (regret pushes the next choice away from the previously chosen option, the free-energy comparison wired into choice) raises the model-generated contrast to +0.08; a counterfactual value learner that updates the foregone option raises it to +0.09 (Table 2). On the second dataset the ordering is the same (+0.09 factual, +0.17 regret bias, +0.16 counterfactual value, human +0.27). Neither counterfactual mechanism improves held-out choice likelihood over the factual learner. So the switching data are evidence that foregone outcomes enter choice, which a factual model under-produces, and not evidence that discriminates our free-energy construction of regret from counterfactual value learning. We class the term as a grounded generative mechanism whose specific form remains to be tested, not as a predictive component.

9 Empirical validation: head-to-head and extension

We tested the model on open data, holding out participants throughout. Provenance, scripts, and full results are in the supplementary `EMPIRICAL_RECORD.md`; Table 3 lists the datasets and what each supports.

Table 2: **Counterfactual switching, data and fitted models.** $P(\text{switch})$ on the next visit to the same state after a loss, split by whether the foregone outcome was better or equally bad; paired t over participants. Held-out NLL per trial: fit on the first 60% of trials, scored on the last 40%. Script `regret_model_fit.py`.

Dataset	Source	foregone better / same	contrast (t)	held-out NLL
Sugawara & Katahira ($n=143$)	human	0.46 / 0.33	+0.13 (10.4)	
	factual Q-learner	0.48 / 0.43	+0.05 (12.3)	0.602
	counterfactual value	0.51 / 0.42	+0.09 (12.4)	0.617
	regret bias (ours)	0.50 / 0.43	+0.08 (9.3)	0.603
Palminteri et al. ($n=20$)	human	0.43 / 0.16	+0.27 (7.3)	
	factual Q-learner	0.39 / 0.30	+0.09 (6.1)	0.354
	counterfactual value	0.44 / 0.28	+0.16 (8.4)	0.368
	regret bias (ours)	0.47 / 0.30	+0.17 (8.6)	0.365

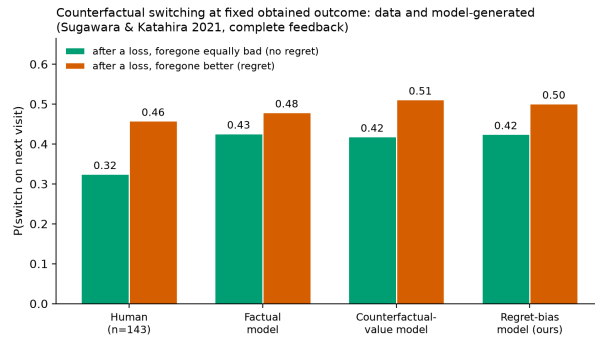


Figure 9: **Counterfactual switching at fixed obtained outcome.** After a loss, people switch more when the foregone option was better than when it was equally bad (Sugawara and Katahira, 2021, $n = 143$). The same contrast generated by three fitted models on the same trials: a factual learner produces less than half of it, the regret-bias and counterfactual-value learners most of it (Table 2).

Table 3: **Datasets.** N is participants (held-out analysis subset in parentheses); ES = effect sizes.

Dataset	N	Signals	Access	Supports
Rutledge GBE happiness (Rutledge et al., 2014)	47,067 (14,803)	gamble choices; momentary happiness	CC0 (Dryad)	subsumption / affect readout
Geschwind–Bringmann ESM (Bringmann et al., 2013; Geschwind et al., 2011)	130 (129)	momentary affect; event pleasantness; worry; baseline neuroticism	CC BY- NC	affect-dynamics; frame-to-worry
Reliability ESM (osf.io/83cfk)	91	momentary affect (12 emotion sliders)	open (OSF)	affect-dynamics replication
Temporal-orientation ESM (Mulholland et al., 2023)	101	per-probe past/future thought; valence	CC BY- NC	direct frame-to-affect coupling
Sugawara & Katahira (Sugawara and Katahira, 2021)	143	choices; obtained + foregone outcomes	open (OSF)	counterfactual signature
Palminteri et al. (Palminteri et al., 2017)	20	choices; obtained + foregone outcomes	open (OSF)	counterfactual replication
Reward+punishment reversal (osf.io/2gq96)	128 (64/64)	learning rates by valence	open (OSF)	general precision deficit
Reward Positivity (MEG) (Pirrung et al., 2025)	90 (52/38)	neural reward response	CC0	reward reactivity (c_{pos})

9.1 Subsumption: our affect layer recovers the happiness equation

We instantiate our three channels over a *gamble* task-model (policies {safe, gamble}; preferences increasing in reward) for the Great Brain Experiment happiness data (Rutledge et al., 2014) (47,067 participants; evaluated held out across a 14,803-participant subset, 96,624 held-out ratings). We reconstruct each predecessor as its single affective channel over this shared task-model (single-channel reconstructions, not the authors’ original implementations) and compare them against the integrated readout on the same held-out ratings (the task-model is Figure 10, its combination with the emotion model is Figure 11; results in Table 4 and Figure 12A). No single channel reproduces the integrated result. The present channel (Pattisapu, RPE) reaches $R^2 = 0.111$ and the forward channel (Hesp, EFE) $R^2 = 0.034$, while the backward channel (Joffily, $-dF/dt$) scores $R^2 = 0.000$. The last is expected rather than a defeat: a rate-of-change signal cannot predict a happiness *level* on a single-shot gamble that has no temporal dynamics to track. The integrated model reaches $R^2 = 0.144$, recovering the standard happiness equation (CR+EV+RPE, $R^2 = 0.146$). No expected-value term is entered as a regressor; expected value instead emerges from the forward/EFE channel, which on this task encodes anticipated reward by construction, and adding it to the present channel raises R^2 from 0.111 to 0.144 (+30%). The happiness equation is thus a special case of our framework.

9.2 Extension: the full model predicts affect dynamics, on two samples

Where the task has temporal structure, the model’s forward dynamics earn their keep, and we report the advantage on two independent experience-sampling samples rather than one. (Which component carries the advantage is the subject of the ablation in §9.2.1; the answer is the valence transition dynamics, not the frame.) In a remitted-depression ESM sample (Bringmann et al.,

factor graph | magenta: outcomes (A) cyan: hidden states (B) blue: controllable puce: policies (G)
red arrows: 1 ascending likelihood, 2 forward, 3 backward messages

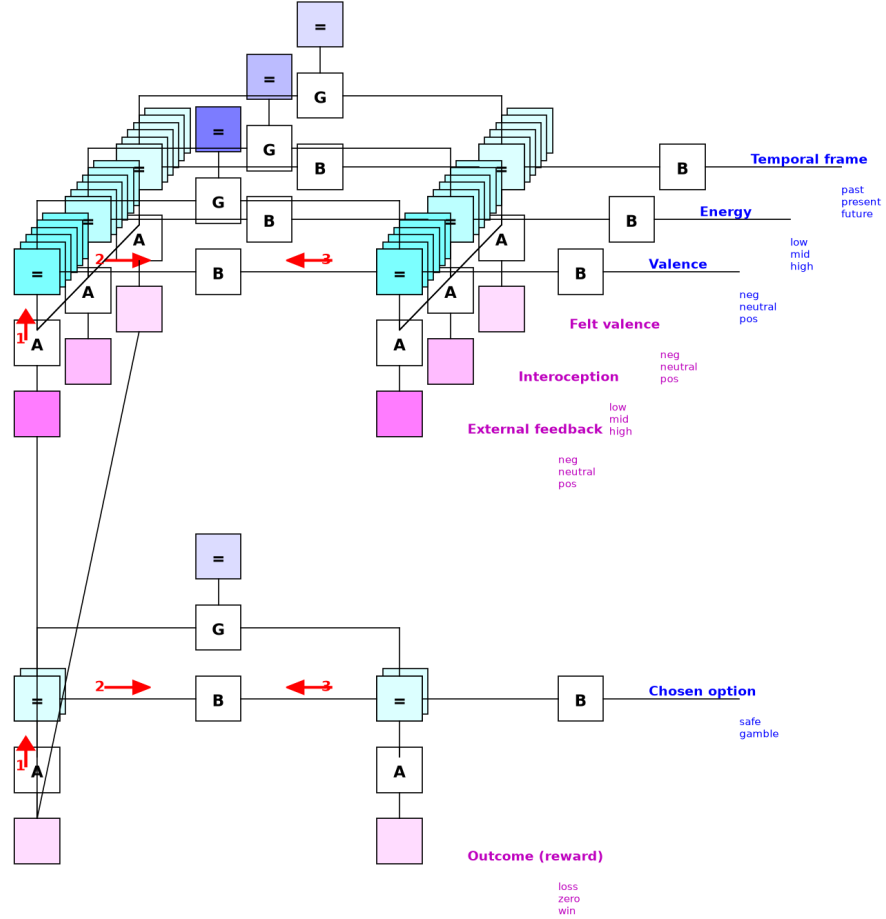


Figure 11: **The gamble task-model joined with the emotion model.** Two-level factor graph of the subsumption test as actually run. The gamble task-model of Figure 10 (lower level: chosen option generating the reward outcome) sits beneath the emotion model of Figure 1 (upper level: valence, energy, and temporal frame with their three modalities). The connecting edges carry the task’s reward outcome into the affect layer, directly as external feedback, and via the three channels reading the task-model’s inference dynamics as the felt-valence observation. Each predecessor model in Table 4 corresponds to retaining a single channel of this upper layer.

Subsumes standard reward-affect models (A); the affect-dynamics advantage is real but sample-dependent: large where affect carries dynamics (B), near parity where affect is already highly persistent (C)

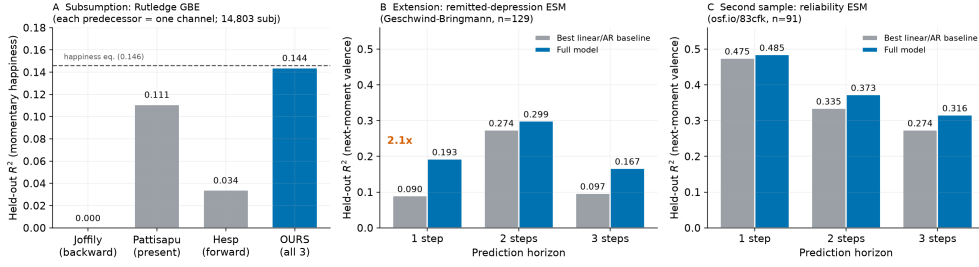


Figure 12: **A general affect-readout framework.** (A) Channel decomposition (Rutledge GBE): each predecessor reconstructed as one channel over the same gamble task-model; the present channel (Pattisapu 0.111) is the informative single-channel comparison, the backward and forward channels being category-mismatched on a static gamble, while integration reaches the reference model (0.144 vs happiness equation 0.146, dashed). (B) On the remitted-depression ESM sample, the full model reaches 2.2 $\times$ the held-out R^2 of the best linear/autoregressive baseline at one step. (C) On an independent reliability ESM sample the model leads at every horizon but by smaller margins, near parity at one step, because affect there is already highly persistent (baseline $R^2 = 0.475$). All held out across participants; Table 6.

2013; Geschwind et al., 2011) (129 participants held out; global parameters fitted on training participants and evaluated on unseen ones), the full model predicts next-moment valence with held-out $R^2 = 0.19$ at one step versus 0.09 for the best linear/autoregressive and event-regression baselines, roughly double the explained variance at this horizon (Figure 12B), with smaller leads at two and three steps. In a second, independent reliability ESM sample (osf.io/83cfk, 91 participants, no event or worry items, so the model runs on valence alone), the lead does not reverse: the model is ahead at two and three steps (participant-bootstrap 95% CIs on the R^2 difference [+0.03, +0.05] and [+0.03, +0.06]) and at parity at one step (+0.01, CI [-0.00, +0.02]; Figure 12C, Table 6). We do not call the one-step result a replication. The two samples differ in exactly the way the framework predicts matters: in the reliability sample affect is already highly persistent (baseline one-step $R^2 = 0.475$ versus 0.089), leaving little non-trivial dynamics for temporal machinery to capture, whereas the clinical sample’s more volatile affect is where the model helps most.

The baselines are: naive persistence; iterated AR(1) ($a + bv_t$, applied h times); direct h -step regression on v_t (the strongest simple predictor at two and three steps); linear event ($a + bv_t + ce_t$) and asymmetric event ($a + bv_t + p[e_t]_+ + n[e_t]_-$), iterated with the mean event beyond the first step. The model’s three global parameters (ρ_{pos} , valence inertia, ω_e) were chosen on an 18-point grid using training participants only; every predictor, model included, receives a train-fit affine calibration.

Component ablations locate the source of the advantage. Driving the model on valence alone in the clinical sample changes almost nothing ($R^2 = 0.20$ at one step), so the gain is not the event channel acting as a hidden regressor. Removing the persistence-like valence-inertia term leaves the clinical-sample advantage intact (the model still reaches $R^2 = 0.14$ at one step and 0.29 at two steps with no inertia at all, versus baselines 0.09 and 0.27); on the already-persistent reliability sample the model sits at parity without inertia (0.46 vs 0.48 at one step, 0.34 vs 0.34 at two), consistent with there being little non-persistence structure to capture there. Where the advantage exists, the model’s own forward dynamics rather than a persistence prior carry it. This holds because the precision-gated prospection operator (§4) keeps forward rollouts calibrated to the agent’s actual affective level rather than drifting toward a fixed optimistic target. We do not claim a head-to-head reimplementations of each predecessor here; the claim is that the integrated model outpredicts standard baselines where affect carries non-trivial dynamics, and single components are insufficient.

9.2.1 Frame ablation: the frame does not carry the prediction

Table 5 isolates the temporal frame. We compare the frame-gated model with the frame-inert version ($g = 0$), with the gated model whose gating posterior is clamped to one frame, and with a regression on the three readout channels plus v_t with no state rollout, all on the same folds with participant-level bootstrap confidence intervals. The result is unambiguous. On both samples and at every horizon the gated, inert and clamped models are within 0.005 of one another in held-out R^2 , so the inferred frame contributes nothing measurable to out-of-sample affect prediction. The advantage over the linear baselines is carried by the valence transition dynamics (the policy-averaged $\mathbf{B}_v$ with inertia), which the channels-only regression lacks: on the clinical sample it trails the full model by 0.08 at one step and 0.05 at three. On the reliability sample the channels-only regression is in fact slightly ahead at one step. The frame’s causal role in this paper is therefore established in the simulated phenotypes (§7), where it changes policy and affect, and not in the experience-sampling prediction, where the observation stream carries no orientation information for it to use.

Table 5: **Frame ablation on both ESM samples.** Held-out R^2 (mean over five participant folds, fold SD in parentheses), paper pipeline and fitted parameters. The last rows give participant-bootstrap 95% CIs (2,000 resamples) for the full model minus the named comparator. Script `esm_frame_ablation.py`.

Predictor	Remitted-depression ($n=129$)			Reliability ($n=91$)		
	1 step	2 steps	3 steps	1 step	2 steps	3 steps
best linear baseline	0.090 (.03)	0.274 (.03)	0.097 (.03)	0.475 (.07)	0.335 (.08)	0.274 (.08)
channels only (no rollout)	0.119 (.07)	0.281 (.03)	0.114 (.05)	0.501 (.07)	0.369 (.08)	0.311 (.08)
full model, frame-gated	0.193 (.03)	0.299 (.03)	0.167 (.03)	0.485 (.07)	0.373 (.08)	0.316 (.08)
frame-inert ($g = 0$)	0.194 (.03)	0.301 (.03)	0.171 (.03)	0.485 (.07)	0.375 (.08)	0.319 (.08)
frame clamped PRESENT	0.194 (.03)	0.300 (.03)	0.168 (.03)	0.485 (.07)	0.373 (.08)	0.316 (.08)
frame clamped PAST	0.193 (.03)	0.299 (.03)	0.167 (.03)	0.484 (.07)	0.372 (.08)	0.314 (.08)
frame clamped FUTURE	0.195 (.03)	0.304 (.03)	0.177 (.03)	0.487 (.07)	0.376 (.08)	0.321 (.08)
full – best baseline, CI	[.09, .12]	[.01, .04]	[.06, .08]	[−.00, .02]	[.03, .05]	[.03, .08]
full – frame-inert, CI	[−.001, −.000]	[−.003, −.001]	[−.006, −.003]	[−.001, .000]	[−.002, −.001]	[−.004, −.001]
full – channels only, CI	[.05, .13]	[.01, .03]	[.04, .08]	[−.02, −.01]	[−.01, .02]	[−.01, .02]

Table 6: **Affect-dynamics prediction on two independent ESM samples.** Held-out R^2 for next-moment valence by prediction horizon, full model versus the best linear/autoregressive baseline on identical inputs, 5-fold cross-validation with whole participants held out. The reliability sample has no event items, so the model runs on valence alone there.

Sample	Predictor	1 step	2 steps	3 steps
Remitted-depression ($n=129$) (Bringmann et al., 2013; Geschwind et al., 2011)	baseline	0.089	0.274	0.097
	full model	0.193	0.299	0.167
Reliability ($n=91$) (osf.io/83cfk)	baseline	0.475	0.335	0.274
	full model	0.485	0.373	0.316

9.3 Latent frame and worry: a null after controls

An earlier version of this paper reported that the posterior future-frame belief covaries with a worry item the model “is never shown” ($r = 0.17$). That was wrong on both counts. The valence composite fed to the model (cheerful and relaxed minus worried, fearful and sad) contains the worry item, and the raw correlation is a between-person valence-level effect. Table 7 reports

the analysis done properly: worry removed from the input, and the association tested after controlling for the model’s own inputs (concurrent valence and event pleasantness), within person, and with participant-clustered standard errors. The adjusted and within-person associations are indistinguishable from zero for the revised model. We therefore make no claim that the latent frame tracks worry. What the frame does is shown in simulation (§7) and in the prediction ablation (§9.2.1); its recovery from affect alone is the identifiability limit discussed in §9.4.

Table 7: **Future-frame posterior vs the worry item, Geschwind ESM** ($n = 11,712$ beeps, 129 participants; script `frame_worry_multilevel.py`). Partial r controls for concurrent valence and event pleasantness; z is the standardised slope over its participant-clustered sandwich standard error.

Model	Worry in input	raw r	partial r	within-person r	z raw	z adjusted	z within
frame-gated (this paper)	yes	0.078	0.006	0.008	6.3	0.5	0.7
frame-gated (this paper)	no	0.067	0.007	0.009	5.9	0.7	0.9
frame-inert (earlier version)	yes	0.142	−0.008	0.028	7.7	−0.6	2.4
frame-inert (earlier version)	no	0.121	−0.003	0.024	6.6	−0.2	2.3

9.4 Direct temporal-orientation test

The strongest test of the temporal-frame construct uses a dataset that labels the orientation of thought itself. In an experience-sampling study with per-probe reports of past- and future-directed thought alongside momentary valence (Mulholland et al., 2023; $n = 101, 1,458$ probes, CC BY-NC), the model’s central structural prediction holds and one of its default assumptions fails. The prediction that past-framing carries negative valence under low positive-belief precision (the RECALL/rumination branch) holds directly and within person: reported past-orientation covaries negatively with momentary valence ($r = -0.23$ within person), and the effect attenuates for people with higher trait positivity ($\beta_{\text{past} \times \text{trait}} = +0.06$), the direction the precision-gated coupling predicts. The default that *future*-framing is optimistic fails: future-orientation is mildly negative in this unselected sample ($r = -0.07$). This motivated the precision-gated FUTURATE operator of §4, in which low precision yields worry-type prospection. Two limits remain. First, the gating softens the negativity of past-framing without reversing it, so the strong reading in which rumination flips into positive narrative stabilisation overstates a real but weaker effect. Second, driven on affect alone the model’s latent frame does not *recover* reported orientation ($r \approx 0$). This is an identifiability limit, not a model failure: felt valence is generated by the valence state, on which the frame acts through the *dynamics* rather than instantaneously, so a single affect reading does not determine the frame; and in the data, momentary affect carries little orientation information in the first place (reported past-orientation explains under 5% of within-person valence variance, $r = -0.23$), which caps any affect-to-frame recovery near that value. The validated claim is therefore a frame-to-affect coupling; recovering orientation would require an orientation-carrying observation channel, not a better inversion of affect.

9.5 What does not improve prediction

Across every dataset we tested, experience-sampling affect, two complete-feedback counterfactual-learning tasks, a probabilistic reward task, and the large-scale happiness data, and for both choice and affect targets, neither the hedonic asymmetry ($c_{\text{pos}} \neq c_{\text{neg}}$) nor counterfactual depth improves prediction. Both are redundant with reward-prediction signals. We therefore present them as *generative* and *neurally/experientially* grounded mechanisms rather than predictive components (Table 8).

Table 8: **What the framework can and cannot claim.** Predictive claims are out-of-sample or out-of-model gains; generative claims are mechanisms that reproduce phenomena and are neurally or experientially grounded but do not improve behavioural prediction.

Claim	Status	Evidence
Multi-channel integration	predictive	recovers the happiness equation (Table 4); outpredicts baselines on two ESM samples, $\sim 2\times$ at one step where affect is volatile (Table 6)
Latent temporal-frame state (worry)	not supported	raw $r = 0.07$ with worry, ≈ 0 after controlling for concurrent valence and events (Table 7)
Latent temporal-frame state (prediction)	see Table 5	frame gating vs frame-inert and frame-clamped variants, participant-bootstrap CIs
Frame-gated dynamics (phenotypes)	generative	RECALL collapse under low ρ_{pos} , future fixation under stress, diathesis-stress attractor (§7); robust across the constant sweep (§7.3)
Counterfactual emotion	generative	regret drives choice-switching ($t = 10.4$); adds no choice or affect prediction
Hedonic asymmetry ($c_{\text{pos}} \neq c_{\text{neg}}$)	generative	blunted Reward Positivity, self-reported anhedonia; not recoverable from choice

10 Clinical mapping: a two-level account of depressive affect

The framework draws a two-level picture of depressive affect and predicts a dissociation between the levels that the data bear out. (i) A *general* reduction in reinforcement-learning rate/precision, present for both reward and punishment learning. In our analysis of an open reward-and-punishment reversal-learning dataset (osf.io/2gq96), depressed participants showed reduced learning rates in *both* conditions (Cohen’s $d \approx -0.5$ to -0.6 , not reward-selective), which maps onto reduced overall precision and replicates the established mapping of anhedonia onto reinforcement-learning parameters (Huys et al., 2013). (ii) A *reward-specific* valuation/reactivity deficit ($c_{\text{pos}} \downarrow$), which converges with reward-specific ventromedial-frontal hypoactivation and blunted Reward Positivity (Pirrung et al., 2025) and with self-reported anhedonia. The two levels dissociate exactly as the model requires: c_{pos} scales the *value* an outcome is assigned, while the learning rate governs how fast beliefs update, so a sample can show reward-specific reactivity blunting (level ii) alongside a general learning-rate deficit (level i) without contradiction, and the Pirrung sample does. The value deficit is what Reward Positivity indexes; the learning deficit is what the reversal task indexes. We label (ii) a model-derived prediction with convergent neural and self-report support that awaits direct behavioural confirmation. A companion paper develops the full bipolar phenotype characterisation.

11 Discussion

11.1 What the integration buys

The single move behind every result here is treating affect as a readout over a task-specific model rather than as a model in its own right. That move has an empirical consequence and a conceptual one. Empirically, the same three channels that recover the happiness equation on a static gamble gain predictive power once the task carries temporal structure, because only then do the valence transition dynamics have work to do; the temporal frame, by contrast, earns its place in the simulated phenotypes and not in the prediction (§9.2.1), which is what

one expects when the observation stream carries no orientation information. Conceptually, prior single-channel accounts stop being rivals and become limiting cases: each is what our readout reduces to when its task offers no dimension for the other channels to read. Where a task lacks temporal structure the framework reduces to the task-appropriate reward-affect model.

11.2 Metastability as health

Health is not a particular temporal configuration but the *capacity to transition* between configurations. The healthy agent visits multiple framing modes with a high switching rate and short run lengths, metastability, in dynamical-systems terms. Pathology is stickiness. Amplified reward sensitivity with impaired interoception locks abstract forward framing on via absorptive narrative inertia; a habit prior locks RECALL on while suppressing EFE-optimal escape. Restoring the ability to transition matters more than correcting the balance between orientations.

11.3 Attention as precision, and mood/emotion timescales

Temporal framing extends attention-as-precision (Feldman and Friston, 2010) to the temporal domain: $\mathbf{B}_{\text{frame}}$ matrices are temporal-attention operators, and framing is deciding *when* to attend. The framework sits at the interface of the mood/emotion timescale hierarchy (Hesp et al., 2021). Individual frame selections are within-trial (M4) phenomena; sustained biases are cross-trial (M5) mood. This is where the thin (affect signals free-energy trajectories) and thick (affect is the enactive force directing attention across horizons) readings of emotion meet.

11.4 Limitations

Emotion-level policy selection uses a one-step counterfactual rollout (Friston et al., 2018); the frame is categorical, not yet a continuous depth; the E-vector is fixed and not yet learned (Watkins, 2008); and the model is single-agent, with dyadic temporal co-regulation (Hinrichs et al., 2025) left to future work. The three valence channels are placed on a common scale by construction (each is a bounded transform of a free-energy quantity in nats, weighted by the frame precisions and then affinely calibrated per dataset), which is a modelling choice rather than a measurement model; a dataset with separate retrospective, present and prospective affect items would let the weights be estimated rather than set. The positive-belief precision ρ_{pos} does three jobs (the identity prior, the RECALL and FUTURATE targets, and the mood-layer state), which is deliberate, one schema parameter in Beck’s sense, but it means the low-mood attractor is built from that coupling; decoupling the three roles and showing which carries the attractor is an ablation we have not run. Empirically, the per-channel head-to-head is on one dataset (Rutledge) and awaits replication; the affect-dynamics advantage holds across two ESM samples with sample-dependent margins and, as §9.2.1 shows, is not carried by the frame. The direct temporal-orientation test (§9.4) validates a frame-to-affect coupling but shows its limits: the model does not invert reported orientation from affect alone (it is not given orientation-carrying input), precision gating attenuates rather than reverses the negativity of past-framing, and the precision-gated FUTURATE operator was introduced after the failed future-optimism prediction and has not been tested on independent data. Grounding the frame in an orientation-carrying observation channel, so the latent frame can be tested for recovery and not only coupling, is the most valuable next step.

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